

How much energy does AI use? There is no way all this AI work being done by everyone and their mom is going to be cheap from a energy perspective.

Posted by u/Nervous_Piccolo_1577 • • 0 points • 15 comments

Powering the training, and application of AI must be massively expensive relative to energy use. Something tells me that in the wide ranging conversation in and around AI, the questions related to AI energy consumption are being left out of the public discourse.

People talk about the impacts on jobs, but perhaps a equally important question is the impact on our energy grid, and in a bigger sense sustainability.

Let's start from the beginning: If I ask AI "are you self aware?" or "can you generate a poem?" then how much energy was consumed to compute such an answer. All I know is, If someone was to ask me to write them a poem, I would need to take a nap afterwards

Comments

u/alexiuss • 1 points • 2023-04-09 14:55 UTC

It's very cheap on more optimized models. I can run smaller models on my desktop just fine.

u/Popular_karma • 6 points • 2023-04-09 14:46 UTC

I can give you a general idea of the energy consumption associated with running large language models like GPT-4.

The energy consumption of AI models depends on several factors, such as the size of the model, the complexity of the task, and the hardware used to run the model. In the case of GPT-4, it has billions of parameters, making it resource-intensive to run. However, the energy consumption for generating individual answers is relatively small, especially when compared to the energy used during the model's training phase.

Typically, the energy usage is measured in terms of FLOPs (floating-point operations per second) or watt-hours. For instance, when running a large language model like GPT-4 on powerful GPUs or TPUs, the energy consumption can range from a few hundred watts to a few kilowatts per hour. However, this energy consumption is spread across many users and tasks, so the energy used for a single answer is a small fraction of that.

It's worth noting that OpenAI and other AI research organizations are actively working on improving the efficiency of AI models, both in terms of computational resources and energy consumption, to make them more sustainable and environmentally friendly.

u/Nervous_Piccolo_1577 • 1 points • 2023-04-09 15:08 UTC

This is great. Thank you

u/oldschoolwitch • 1 points • 2023-04-10 14:17 UTC

You could have just got this from chatGPT yourself :)

u/Automatic_Tea_56 • 1 points • 2023-04-09 13:18 UTC

Maybe you should ask what type of energy as well. Many large companies are covering with renewables.

u/L2ne • -2 points • 2023-04-09 11:54 UTC

long answer short: your stupid!

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Model Prompt Completion

8K context \$0.03 / 1K tokens \$0.06 / 1K tokens

32K context \$0.06 / 1K tokens \$0.12 / 1K tokens

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if they can sell it for that cheap then it gonna cost them even less in energy as they gonna want to make a profit ...

u/Nervous_Piccolo_1577 • 1 points • 2023-04-09 13:00 UTC

Bro. You are not thinking about this correctly. This about the training for each model. Based on what I have read chatgpt is built on hundreds of thousands of models.

All of this processing is not free, and is currently not cheap, nor do i think it ever will be.

Every time the AI does work it is processing information and that takes energy. A lot of energy

u/L2ne • 1 points • 2023-04-10 11:20 UTC

are you high ? they sell a product for x wich implies the cost for the product is x- something ... they sell you 1k tokens for 0.06\$ or 0.12\$ while 1k tokes are approximately 750 words. so you ask chat gpt a 750 word question and it cost you 12 cents ... not that epensive i think...

maybe im not getting what you really mean but to me it feels like you fear monger about the energy consumed while not looking at the stats.

u/IndianaHorrscht • 1 points • 2023-04-09 12:59 UTC

You're

u/GreenRiverSuccess • 0 points • 2023-05-21 23:05 UTC

Look at this edge lord being grammar police

u/TikiTDO • 3 points • 2023-04-09 12:39 UTC

8k is like a few paragraphs of text, and it will run you around 15 to 30 cents per decently large query. A kwh costs on the order of 15 to 30 cents. Obviously an answer takes less than an hour, and these servers will usually peak at a few kw of power draw.

In what world is that "cheap"?

u/Nervous_Piccolo_1577 • 2 points • 2023-04-09 13:07 UTC

My next query: Please answer the following: How many queries are there per day, and what would that amount to in terms of kwh world wide? Compare that number to other industries and determine an analogy to paint a picture for basic human (above) to understand conceptually how big of an energy hog AI is and what it will become in the future”

u/TikiTDO • 1 points • 2023-04-09 13:16 UTC

You seem to have confused Reddit and an AI chat prompt. You might want to ask gpt or Bing that

u/rydan • 3 points • 2023-04-09 08:41 UTC

In the longterm AI actually uses negative energy. Right now you have to frontload all of it using coal, nuclear, or whatever. But in the future it will help us reach the singularity. Then it will pay back the energy cost by inventing fusion for us.

u/ReboundRecruiting • 1 points • 2023-04-09 20:13 UTC

We've already invented fusion it's just not establishable as a net positive for energy yet